

Marketing Section (LPG and POL)

Marketing & Operations (MO)

Marketing Operation group commenced functioning in 1986 i.e. Since inception of Oil Industry Safety Directorate under the Ministry of Petroleum & Natural Gas.

Marketing Operations involve management of the supply chain ensuring that the petroleum production is met on time at the most optimum placement cost all across the country while maintaining the highest standards of safety. Towards this end, supply chain nodes had been set up to link the refineries with the consumption centres via multi-modal transport systems like Pipelines, Railway Tank Wagons and Road Tank Trucks. These nodes include Terminal / Depots, Aviation Fuelling Stations, LPG Bottling Plants, Lube Blending / Small Can Filling Plants, Port based Import/ Export facility, Refrigerated LPG storage facilities etc. These locations receive products / inputs in bulk through the transport mode/s mentioned above; keep them in safe custody and transfer them to market friendly packages through specially designed filling systems. Most of these nodes are equipped with automated systems ensuring fail-safe, efficient and user-friendly operations.

The typical operating blocks of POL installations are:

1. Tank Farm
2. Cross Country Pipeline custody transfer point
3. Tank Wagon / Tank Truck Loading / Unloading Facilities
4. Product Pump House/ Air Compressor House
5. Fire Pump House, Fire Hydrant & other firefighting facilities
6. Quality Control Lab.
7. Electrical systems including Incoming, Distribution, Lightning protection and PMCC
8. Control Room for TFMS, LRCS (in case of automated locations), S&D functions and ERP systems
9. Cathodic Protection systems
10. Storm water drainage / Effluent collection, treatment, OWS and disposal system.
11. Vapour Recovery Systems

Similarly, the typical operating blocks of LPG Bottling Plants are:

1. Bulk Storage Vessels
2. Cross Country Pipeline custody transfer point
3. LPG Bulk Tank Wagon / Tank Truck loading/ unloading facilities
4. Cylinder Filling Shed
5. Cylinder Storage Shed
6. Product Pump / Compressor and air compressor house
7. Cylinder evacuation & Valve change facility
8. Fire Pump House, Fire Hydrant & other firefighting facilities
9. Electrical systems including Incoming, Distribution and PMCC
10. Storm water drainage / Effluent collection, treatment and disposal system.
11. Control room with HC detection and Fire alarm annunciation system

12. Cathodic protection systems

Key differentiating features of Marketing Operations systems (as against Refineries) are:

1. Operation in discrete shifts (as opposed to 24X7 operating style of Refineries / Pipelines)
2. Variation in workload in tune with market demand.
3. Ingress of third-party vehicles & personnel and engagement of contractors' labour all through the working shifts, thereby increasing process variability.
4. Unavailability of regular employees in non-working shifts.
5. Human habitation in proximity of the locations with nominal buffer space.
6. Absence of active source of ignition within location's licensed area.

Various activities are carried out by this group viz. Safety Audits (ESA/SSA/PCSA), standard development and revision, Investigation of accident/incident and Knowledge dissemination to maintain high degree of safety in the Oil Industry.

Out of all the Standards (STD)/ Guidelines (GDN)/ Recommended Practices (RP) of OISD, the following critically pertain to marketing operations; the list, needless to mention, is only indicative and not exhaustive. Critical issues covered in other relevant OISD STD/GDN/RPs e.g. OISD-STD-105 (Work Permit), OISD-STD-110 (Static Electricity), OISD-GDN-115 (Fire Fighting Equipment), OISD-STD-118 (Layout), OISD-STD-137 (Insp. of Elec. Equipment), OISD-STD-142 (Insp. of firefighting Equipment), OISD-STD-145 (Internal Safety Audit), OISD-STD-178 (Management of Change), OISD-GDN-206 (Safety Management System) remain applicable for marketing operations as well (in addition to other type of locations included in the scope of the said standards). Users are strongly advised to refer to ALL, repeat, ALL OISD Standards / Guidelines / Recommended practice before arriving at any conclusion.

Standards pertaining to Marketing operations:

A. Design, Layout & Operating Practices

S.No.	Standard No.	Standard Name
1	OISD-STD-118	Layouts for Oil & Gas Installations
2	OISD-STD-144	Liquefied Petroleum Gas (LPG) Installations
3	OISD-STD-150	Design and Safety Requirements for Liquefied petroleum Gas Mounded Storage Facility.
4	OISD-STD-159	LPG Tank Trucks: Requirements of Safety on Design / Fabrication and Fittings
5	OISD-GDN-169	OISD Guidelines on Small LPG Bottling Plants (Design and Fire protection facilities)
6	OISD-STD-210	Storage, Handling and Refueling of LPG for Automotive use
7	OISD-STD-225	Storage, Handling & Dispensing at Petroleum Retail Outlets

8	OISD-STD-235	Storage, Handling, Refuelling and fire fighting at aviation fuelling stations
9	OISD-STD-236	Design, Layout, Operation & Maintenance of Refrigerated LPG Storage
10	OISD-STD-237	Layout, Design consideration, Safety, Operation and Maintenance of Lube/ Grease manufacturing and filling plants
11	OISD-STD-244	Storage and handling of Petroleum Products at Depot and Terminals including Standalone Crude Oil Storage Facilities

B. Training

S.No.	Standard No.	Standard Name
1	OISD-STD-154	Training in midstream & downstream sector of petroleum industry

C. Safety & Fire Protection

S.No.	Standard No.	Standard Name
1	OISD-STD-105	Work Permit System
2	OISD-STD-117	Fire Protection Facilities for Petroleum Depots, Terminals Pipelines Installations and Lube Installations
3	OISD-STD-156	Fire Protection Facilities for Ports Handling Hydrocarbons
4	OISD-RP-157	Safety in Transportation of Bulk Petroleum Products (Rail and Road)
5	OISD-STD-162	Safety in Installations and maintenance of LPG Cylinders Manifold
6	OISD-STD-173	Fire Prevention and Protection System for Electrical Installations
7	OISD-GDN-192	Safety Practices during Construction
8	OISD-GDN-206	Guidelines on Safety Management System in Petroleum Industry
9	OISD-STD-211	Safety in Petroleum laboratories

D. Maintenance & Inspection:

S.No.	Standard No.	Standard Name
1	OISD-STD-128	Inspection of Unfired Pressure Vessels
2	OISD-STD-129	Inspection of Storage Tanks
3	OISD-STD-130	Inspection of Piping Systems
4	OISD-STD-132	Inspection of Pressure Relieving devices

5	OISD-STD-135	Inspection of Loading & Unloading hoses for Petroleum Products
6	OISD-STD-137	Inspection of Electrical Equipment
7	OISD-STD-142	Inspection of fire fighting equipments and systems
8	OISD-RP-147	Inspection & safe practices during electrical installations
9	OISD-RP-149	Design aspects for safety in electrical systems
10	OISD-GDN-233	Guidelines on inspection of non-piggable pipelines.
11	OISD-STD-240	Asset Integrity Management of rotary equipment

E. Other Applicable Standards

S.No.	Standard No.	Standard Name
1	OISD-RP-110	Recommended Practices on Static Electricity
2	OISD-STD-113	Classification of Area for Electrical Installations at Hydrocarbon Processing & Handling Facilities
3	OISD-GDN-115	Guidelines on specification of fire fighting equipment & mitigation strategy for credible fire scenarios
4	OISD-STD-141	Cross Country liquid hydrocarbon pipeline
5	OISD-STD-145	Internal Safety Audits
6	OISD-STD-155	Personal Protective Equipment
7	OISD-GDN-161	LPG Tank Truck Incident : Rescue & Relief Operations
8	OISD-STD-163	Safety of Control Room for Hydrocarbon industry
9	OISD-GDN-165	Guidelines for Rescue & Relief Operations for POL Tank Truck Accident
10	OISD-RP-167	POL Tank Lorry Design & Safety
11	OISD-STD-178	Management of Change
12	OISD-GDN-180	Lightning Protection
13	OISD-GDN-207	Contractor Safety
14	OISD-STD-227	Emergency Response and Preparedness in Oil & Gas Industry

The important documents referred to in the context of safety in Marketing Operations locations include:

- HIRA Study report – findings and implementation.
- PESO approved drawings
- Emergency Response & Disaster Management Plans.

Although the overall status of all OISD STD/ GDN /RPs are advisory in nature, 9 OISD Standards (indicated below) had become mandatory in Marketing Operations area by virtue of having been included in the following statutory regulations:

- Petroleum Rules, 2002
- Gas Cylinder Rules, 2016

- Static & Mobile Pressure Vessels (Unfired) Rules, 2016

List of 6 Nos. Standards included in Petroleum Rules, 2002

S.No.	Standard No.	Standard Name
1	OISD-STD-105	Work Permit System
2	OISD-STD-117	Fire Protection Facilities for Petroleum Depots, Terminals, Pipeline Installations & Lube Oil Installations
3	OISD-STD-118	Layouts for Oil & Gas Installations
4	OISD-STD-141	Design & Construction Requirements for Cross Country Hydrocarbon Pipelines
5	OISD-STD-156	Fire Protection Facilities for Ports Handling Hydrocarbons

List of 2 Nos. Standards included in Gas Cylinder Rules, 2016

S.No.	Standard No.	Standard Name
1	OISD-STD-144	Liquefied Petroleum Gas (LPG) Installations
2	OISD-STD-169	OISD Guidelines on Small LPG Bottling Plants (Design and Fire Protection Facilities)

Standard included in The Mobile Pressure Vessels (Unfired) Rules, 2016.

S.No.	Standard No.	Standard Name
1	OISD-STD-150	Design and Safety Requirements for Liquefied petroleum Gas Mounded Storage Facility